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Imagine this: Memories of fiction are used in mental simulations in the absence of lived experience

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Abstract
Memories of events from fictional sources (e.g., scenes from movies or novels) share many properties with memories of lived experiences (Yang et al., Journal of Experimental Psychology: General, 151 (5), 1089, 2022). Here we test whether memories of fictional events can serve a similar function to personal memories, serving as the building blocks for simulations of novel scenarios. Across two studies, participants imagined themselves in future scenarios, rated the phenomenological qualities of each simulation (e.g., visual properties), and then identified the sources (e.g., fiction, lived experience) used to generate their simulations. Study 1 ($N = 208$) focused on people's simulations of a pandemic as a function of whether they had personally experienced the COVID-19 lockdown yet. Participants tested prior to lockdown reported drawing on fictional sources when simulating a pandemic scenario; in contrast, participants tested after lockdown began drew on their own lived experiences when simulating. Study 2 ($N = 248$) replicated these results using a diverse set of scenarios (e.g., being stuck in an elevator) selected to elicit different levels of prior experience across participants. Again, fictional memories were relied upon when simulating in the absence of lived experience. The results suggest that fictional and personally experienced memories can serve similar functions. Theoretically, the results are consistent with arguments to broaden frameworks of autobiographical memory to include memories of fictional events, in the same way that false memories and vicarious memories are considered to be types of autobiographical memories.

Keywords Fictional memory · Episodic simulation · Autobiographical memory

Introduction

Imagine driving down the road and seeing a car pulled over on the dirt shoulder, with a woman inside, slumped over the steering wheel. How would you decide what to do next? In 2019, a 21-year-old auto technician from Arizona encountered this very situation (Kreps, 2019). Despite having no formal CPR (cardiopulmonary resuscitation) training, he remembered and applied a scene from the sitcom *The Office*, which highlighted that the beat "Stayin' Alive" by the Bee Gees has the ideal tempo for chest compressions (Silverman & Lieberstein, 2009). The woman started breathing, and emergency responders arrived 10 min later. This situation illustrates our focus here: We explore the idea that people may draw on specific event memories from movies, novels, and fictional sources in situations they lack lived experience with, even while correctly identifying the source of the memory as fictional.

Humans often think about the future – from the mundane (e.g., anticipating the taste of coffee) to upcoming challenges and successes (e.g., imagining crossing the finish line of a marathon; Szpunar et al., 2014). The ability to think about the future is an essential component of human cognition; it allows us to utilize information from our past experiences to simulate, predict, plan, and optimize future goals and decisions (Szpunar & Schacter, 2013; Szpunar et al., 2014). There is a consensus among memory researchers that such simulations draw on the same neural and psychological processes as does remembering the past (Atance & O'Neill, 2001; Schacter & Addis, 2007; Addis et al., 2010; De Brigard et al., 2016; Schacter et al., 2017). That is, we

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flexibly retrieve and reconstruct elements of our episodic memory to build novel simulations of the future (i.e., *constructive episodic simulation hypothesis*; Schacter & Addis, 2007). Behavioral, patient, and neuroimaging data support this claim. For example, it is easier to simulate a scenario in a familiar location (i.e., with strong pre-existing memory representations) than a location one has never visited (e.g., the pyramids; Arnold et al., 2011; Robin & Moscovitch, 2014). Similarly, Szpunar and McDermott (2008) found that simulations situated in recently experienced places (e.g., university settings for undergraduates) were more vivid and sensory-rich than simulations imagined in contexts experienced in the more distant past (e.g., high school). Patient data give more direct evidence for an overlap in brain regions underlying remembering and imagining the future. Amnesics, for example, are impaired in their ability to imagine the future, in addition to their well-documented deficits in episodic memory (Klein et al., 2002; Kwan et al., 2010; Prigatano et al., 1993). Neuroimaging provides additional evidence; a set of brain regions including the medial temporal and frontal lobes, posterior cingulate and retrosplenial cortex, and lateral parietal and temporal areas show similar patterns of activation regardless of whether participants are cued to remember a past event or to imagine a future event (e.g., Addis et al., 2007; Szpunar et al., 2007). This group of brain regions, referred to as the default network, appears to be particularly important for constructing and reconstructing visuospatial context, a key component of both memories and simulations of the future (Cabeza & Nyberg, 2000).

In contrast to the strong evidence for overlap between remembering and simulating events (Buckner & Carroll, 2007; Schacter et al., 2012), less is known about the specific building blocks used to construct episodic simulations of the future. Prior literature has focused primarily on personal event memories as the foundations for simulations. However, people often have many relevant experiences stored in memory, raising the question of why some memories are selected over others. Recent work by Bordalo et al. (2025) addressed this experimentally by manipulating the availability of memories at the time of simulation: participants were primed to retrieve one of two types of memories (e.g., identity theft vs. loss of loved one) before estimating the risk of a cyberattack. Critically, the prime did not change *what* was stored in memory but rather affected the availability of particular memories; recently retrieving a memory of a similar crime (i.e., identity theft) boosted participants' risk estimates of a cyberattack. In the *General discussion*, we revisit the issue of competition between memories during the simulation process.

Here we suggest that memories of fictional events can come to mind when simulating future events, especially when people lack direct experience with the simulation target (i.e., when they do not have memories of lived experience to draw from). By memories of fictional events, we mean memories of specific events from works of fiction (e.g., specific scenes from novels, movies, TV shows). There is ongoing debate in the literature about whether memories of fiction are fundamentally distinct from memories of lived experience. Gander et al. (2025), for instance, argue that fictionality is a core dimension of memory, suggesting that fictional memories constitute a distinct type of memory. In contrast, other researchers argue that memories of fiction and lived experience are not fundamentally different types of memory (Semeijn, 2024). While it may seem surprising to equate fictional events with lived experience, memory researchers have a long history of using movie scenes and short stories as laboratory analogs of episodic memory. Individuals easily retrieve specific memories of fictional events, regardless of whether the cues are general prompts or random words (i.e., a modified Galton-Crovitz word-cueing technique, Crovitz & Schiffman, 1974; Galton, 1880; Yang et al., 2022). For example, in one study, participants retrieved a "frequently talked about memory" and rated its properties on items adapted from the Autobiographical Memory Questionnaire (Rubin et al., 2003) such as feelings of reliving, auditory and visual imagery. Critically, some participants retrieved memories from fictional works and others retrieved personal memories. Further, while personal memories were overall more vivid than memories of fiction, personal and fictional memories showed similar patterns in phenomenological and memory function ratings (Yang et al., 2022).

While there is not a clear consensus among researchers on how important the fact-fiction distinction is in memory (see Gander et al., 2023, for an argument that this is a core psychological dimension), what is key for present purposes is that participants report retrieving personal and fictional memories for similar reasons, as measured using a modified version of the Talking About Life Experiences questionnaire (TALE; Bluck et al., 2005, as modified by Yang et al., 2022). The TALE measures three possible uses of memories: *directive* (e.g., "My memory of this event impacts my life decisions"), *social* ("My memory of this event influences the relationships I have with others"), and *self* ("My memory of this event helps me to better understand myself"). Here our focus is on the directive function of memories, and how memories can be retrieved to direct future actions – just like when the auto technician drew on a scene from *The Office* to perform CPR on an unconscious woman. Specifically, we explore whether fictional memories are particularly likely to be used in episodic simulations of the future when people have no memories of personal experience on which to draw.

Across two studies, we asked people to simulate specific scenarios that varied in their likelihood of cueing memories of related lived experience (i.e., first-hand personal experience). That is, while none of the to-be-simulated

scenarios were expected to have been experienced in whole, we expected them to vary in how much and how close participants' prior experiences were to the scenario.

In Study 1, scenarios were originally designed such that one (e.g., 1 st day of university as a transfer student) was expected to cue more related memories of lived experience whereas the others (e.g., being stranded on a deserted island) were expected to cue fewer memories (or less related ones). Critically, one of the latter scenarios was a pandemic-related event, and by some grim twist of fortune, the COVID-19 pandemic began mid-data collection, providing a naturalistic opportunity to examine participants' simulations as they gained relevant lived experience in real time. We shifted our main analyses to a within-item analysis comparing responses to the pandemic scenario before and after the onset of the COVID-19 lockdown, hypothesizing that participants would have more memories of relevant lived experiences to draw on in the latter case. In Study 2, pilot testing was used to develop a list of general events (e.g., being in a limo) that people varied naturally in their experience with. For each general event, a specific to-be-simulated scenario was created (e.g., *Imagine you are getting picked up by a car service to go to the airport. When you go outside, you are surprised to see it is a limousine.*) In the experiment, each participant reported how much lived experience they had with a list of general events. Based on these ratings, each participant was assigned four specific to-be-simulated scenarios: two *high-experience* scenarios (general experience rating ≥ 4) and two *low-experience* scenarios (general experience rating ≤ 3). To preview, across both studies, people drew on both memories of fiction and lived experience when simulating new events; critically, people drew on memories of fiction in the absence of relevant lived experience.

Study 1

In Study 1, we report the results of a natural experiment exploring the relationship between individuals' lived experiences and their propensity to use fiction as a source for their mental simulations.

Methods

Data and materials are available online on the Open Science Framework (https://osf.io/hxqck).

Participants

We recruited 238 participants (150 women, 86 men, one non-binary gender, one non-disclosed; $M_{age} = 19.20$ years, $M_{SD} = 1.40$, $n_{before} = 95$, $n_{after} = 143$) from an undergraduate participant pool between December 2019 and April 2020, who participated for course credit. The sample was 47.1% freshmen, 37.5% sophomores, 10.6% juniors, 4.33% seniors. Qualitative data were also collected from 30 additional pilot participants: these data are excluded in the present analyses.

Materials

During the fall of 2019, we developed and piloted four items designed to investigate the relationship between narrative fiction and people's simulations. These items included three general events that our undergrad participant population were less likely to have directly experienced and one they were more likely to have direct experience with. One of these items was a pandemic scenario that asked participants to, "*Imagine that a highly infectious disease is quickly ravaging the globe, inciting panic and destabilizing societies.*" At the time, this scenario was chosen because it was *not* expected to cue direct memories of lived experience, however, partway through data collection, the world took an unexpected turn as the COVID-19 pandemic began to unfold. This allowed us to investigate the effect of lived experience on participants' simulations of this pandemic scenario while holding constant the fictional source material available. That is, while the fictional source material available to our participants did not change dramatically in the 1 year that the data were collected, participants' life experience with a pandemic and lockdowns shifted dramatically mid-data collection, allowing an unforeseen environmental manipulation after the COVID-19 pandemic fundamentally disrupted students' lives.

Procedure

There were three main phases to this study: (1) *the simulation phase*, (2) *the reporting phase,* and (3) *the experience assessment phase*.

In the *simulation phase*, participants were presented with four specific scenarios (see Appendix A). Participants were asked to vividly imagine and describe "what the experience would be like and what you would do" in one to three sentences (these descriptions were later used to cue the simulations during the reporting phase). All four scenarios were simulated before participants moved to the reporting phase to reduce the chances that subsequent questions about source material or vivacity would affect the contents of what participants imagined.

In the *reporting phase*, participants answered questions about each of their simulations, with items selected from the modified version of the Autobiographical Memory Questionnaire used in Yang et al. (2022). Participants were reminded of what they wrote for each scenario and answered Autobiographical Memory Questionnaire (AMQ) questions relating to visual and auditory vivacity, a sense of reliving,

emotional intensity, spatial layout (see Appendix B for a list of AMQ items). Participants were then asked to report what the "sources or material" were for their simulations as specifically as possible. They were told:

> You might have used one or many sources. The sources could be from anywhere, including but not limited to: your past experiences, experiences from other people, works of fiction, videos you've seen, dreams, or anything else. It is likely you will not be able to remember all of the sources you used for your simulation. Just try your best!

Participants first responded in as much or little detail as they desired in a text box, including, "I'm not sure." On the next screen, participants categorized the source(s) (for the same simulation) using a multi-select checklist, selecting all that applied from the following categories: lived experience (first-hand experiences), vicarious experience (experiences heard about from another person), nonfiction (read or watched), and fiction (read or watched). If participants indicated drawing upon fiction, they were asked follow-up questions about the subcategory (e.g., book, short story, movie, TV show).

In the *experience assessment phase*, participants reported how much lived experience they had with the general event related to each of the specific scenarios (1 = no experience, 6 = a great deal of experience). For example, they were asked how much experience they had with "infectious diseases" (see Appendix A for a full list of general events).

Statistical analyses

Analyses were completed in RStudio (2025) running R version 4.4.1. Binomial logistic regressions were run using the "lme4" software package.

Results

The original analysis plan was to compare the phenomenology and source reports for simulations relating to scenarios that we expected people to have less direct experience with (pandemic, deserted island, undercover in prison) versus a scenario we expected to be closer to participants' prior experiences (first day of school at new university). However, as mentioned above, the COVID-19 pandemic provided us with a unique opportunity to switch from a between-item analysis to a within-item analysis. That is, we compared undergraduate participants' simulations of a pandemic as a function of whether their data were collected prior to their campus lockdown (10 March 2020) versus after. The Online Supplementary Material (OSM) contains the results of the original planned analysis, excluding the pandemic item as it could no longer be assumed to reflect low lived experience for the majority of participants. The results in the OSM parallel those reported here.

We created a *COVID-19 campus lockdown experience* variable to capture participants' transformed level of experience with a global pandemic, separating participants who completed the study before and after 10 March 2020 (inclusive). This date was chosen because it was the date when the participants' university formally announced that the remainder of the academic semester would be remote, serving as a clear event boundary for our undergraduate participant population. This announcement occurred while students were at home for Spring break, highlighting the difference between pre and post 10 March for these students – they left for Spring break and did not return for the rest of the semester. As anticipated and shown in Fig. 1, the two groups reported different levels of direct experience with infectious diseases (1 = no experience, 6 = a great deal). Participants who responded prior to campus lockdown reported less experience with infectious diseases ($M_{before} = 2.20$) than those who responded on or after campus lockdown ($M_{after} = 5.08$), $t(206) = 15.40$, $p < .0001$, $n_{before} = 66$, $n_{after} = 142$.[1]

Content of simulations

The simulations collected before versus after the campus lockdown (10 March 2020) were qualitatively different, as illustrated in the sample responses in Table 1.

These subjective impressions were quantified through an analysis of participants' ratings of their simulations, as reported in response to items from the AMQ. That is, we explored whether simulations differed in their vividness (visual imagery, auditory imagery, spatial imagery), intensity of emotions, and sense of reliving as a function of COVID-19 lockdown experience. We computed a multivariate analysis of variance (MANOVA) to examine the effects of lockdown experience (before vs. after 10 March) on AMQ items. There was a statistically significant effect of COVID-19 lockdown experience: Pillai's trace =.25, $F(5, 232) = 15.57$, $p < .001$, indicating that AMQ characteristics differed depending on whether participants completed the study before or after their college education was directly impacted by COVID-19. Simulations created after campus lockdown were rated an estimated 1.00 point (7-point scale) higher than those created prior to lockdown, $SE = 0.20$, $t = -5.68$, $p < .0001$. In other words, episodic simulations of the pandemic were more vivid overall when lived experience

[1] Predicting reported experience for the pandemic scenario in a linear regression with date as continuous predictor yields the same conclusion: $F(1,206) = 177.7$, $p < .0001$. Data were not collected continuously across time (but rather in "buckets" as slots became open in an internal study management system); thus, we opted for the more categorical approach in primary analyses.

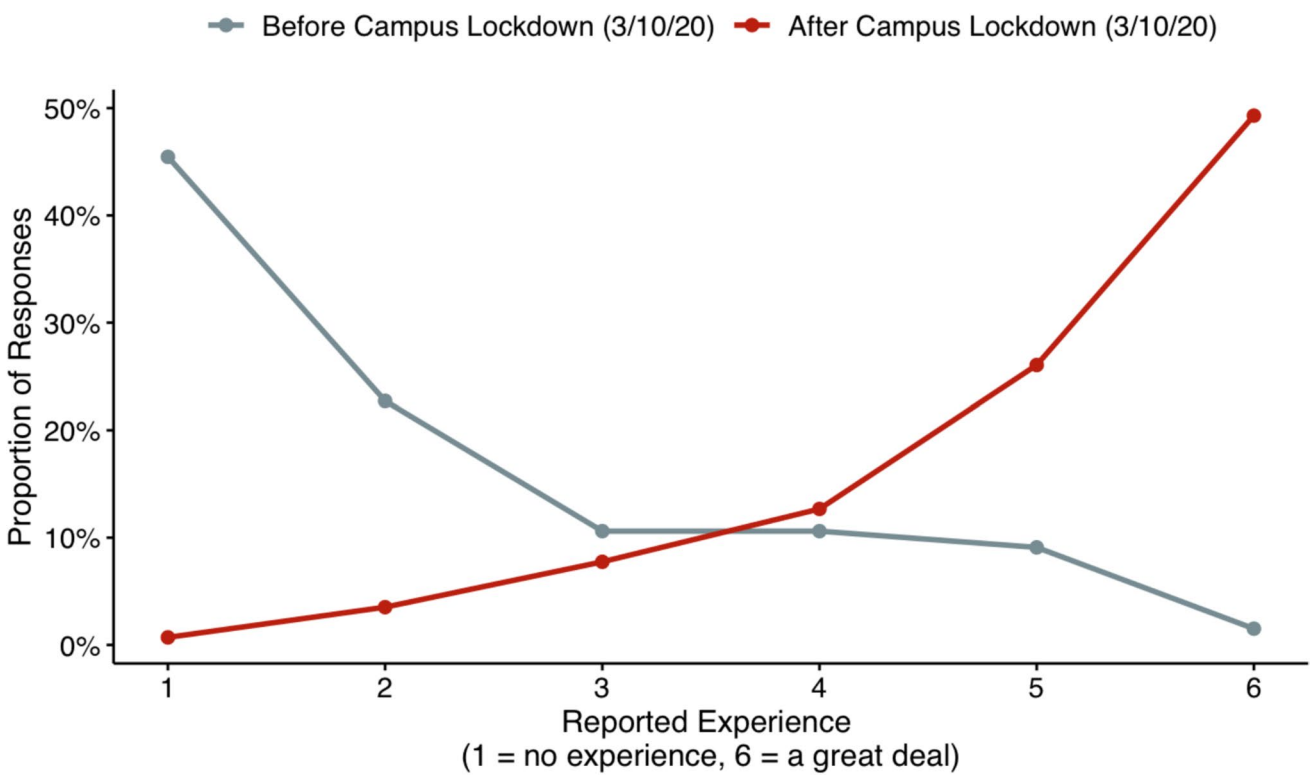


Fig. 1 Distribution of reported experience with infectious diseases by COVID-19 lockdown experience

with the COVID-19 lockdown was available as source material (see Fig. 2).

Sources of simulations

We found consistent evidence that participants recruited memories from fictional sources when simulating specific scenarios. An initial scoring of participants' open-ended responses when asked to report the source(s) of their simulations (see OSM) suggested that fiction was reported as a source at similar rates regardless of whether participants made open-ended reports or used the checklist. Thus, for parsimony we report only the checklist data; critically, the sources were not mutually exclusive: participants could select as many of the categories as they believed applied. Since "read" sources were rarely selected (and because we had no hypotheses about modality), we collapsed over modality in our analyses. That is, the analyses collapsed "read fiction" and "watched fiction" into "fiction" and "read non-fiction" and "watched non-fiction" into "non-fiction." Thus, each simulation was coded for four source variables (i.e., fiction, lived experience, non-fiction, vicarious), with a 1 indicating that the source was selected from the checklist and a 0 meaning it was not selected. For example, a simulation for which a participant selected fiction and non-fiction would be coded as 1 for fiction, 1 for non-fiction, 0 for lived experience, and 0 for vicarious. This approach allowed us to examine the probability of selecting each source independently of the others (see Table 2). Full descriptives are reported in Table 2 and each source type (fiction, lived experience, non-fiction, vicarious) was modeled as a separate binary outcome variable in logistic regressions. However, given that our goal was to test a targeted prediction about the role of fiction in episodic simulation, we report here the analyses predicting the inclusion of fiction and lived experience as source material. The analyses predicting reliance on non-fiction and vicarious sources are included in the OSM; to summarize, experience was unrelated to reliance on non-fiction sources but somewhat positively related to usage of vicarious memories (points we return to in the *General discussion*, given that the patterns differed in Study 2).

Our key predictions involved two binomial logistic regressions. In the first, individuals' reported ratings of direct experience with infectious diseases (1 = no direct experience, 6 = a great deal of direct experience) were used to predict the likelihood that fictional materials were reported as source materials for the pandemic simulation. In the second, reported experience with infectious diseases (1–6) was used to predict the likelihood that memories of lived experience were reported as source materials for the pandemic simulation.

Table 1 Sample participant responses of pandemic simulations

Date	Description of simulation	Description of source(s)
12/02/2019	"I imagined myself locked in my house with my family, watching the TV with the lights off. I am texting my boyfriend worriedly, as he is with his own family and not with me. I feel panicked and terrified in the scenario and am praying often that the disease does not touch anyone I love."	"I watched a movie as a part of my eighth-grade science class, called *Contagion*, that was so harrowing it has stuck with me all these years. To this day, I am terrified of being patient zero. Most of the fear and hopelessness that the scenario dredged up for me has roots in this movie. Additionally, I am a huge fan of *The Walking Dead*, and the visualization I formed of the world outside my house was very similar to images of the world portrayed in this show."
12/3/2019	"Total anarchy. Cities burning, people rioting in the streets, a total collapse of society. People dying everywhere from disease."	"Disaster fiction like the movie *Pandemic* or the *Book World War Z*. The mobile game *Plague Inc* in which you infect the entire world as a disease."
1/24/2020	"I'm worried for my family, especially for my son. I will keep my family from going to a public place. I am doing research online to see which countries are relatively safer and plan to move with my family."	"I thought about what I would do. I would definitely put my family first. I've had a similar experience when SARS broke out in China."
3/5/2020	"I am imagining what is going on now with the coronavirus. Some people are too worried, some not enough and some are making a joke about the whole thing."	"This simulation came from what is actually happening right now. I think another source for this would be news outlets."
3/12/2020	"I imagine myself right now laying in my bed depressed about school getting canceled"	"From my real life."
3/28/2020	"The streets are empty. I am staying indoors all the time and feeling bored. My parents are panicking."	"What is happening currently. (In fact, I can barely imagine this one because it's so real.)"

Table 3 provides the details of the model using reported experience (1 = no direct experience, 6 = a great deal of direct experience) to predict the likelihood of using *fictional* source material as a basis for one's simulation. Notably, the odds ratio was.50. In other words, for each 1-unit increase in reported experience on the 1–6 scale, there was a 50% decrease in the odds of reporting that a work of fiction was recruited as part of the simulation.

The second binomial logistic regression indicated that higher ratings for direct experience with infectious diseases (1 = no experience, 6 = a great deal) predicted an increased likelihood of reporting memories of lived experience as source material when simulating a specific pandemic scenario. Specifically, for each one-unit increase in reported experience, the odds of recruiting lived experience into one's simulation almost tripled (OR = 2.66, 95% CI [2.07, 3.57], p <.0001, see Table 4).

Taken together, these two models are consistent with the hypothesis that people are more likely to recruit material from works of fiction into their simulations in the absence of relevant lived experience. This is illustrated in Fig. 3.

Study 2

Study 1 found a clear relationship between participants' self-reported prior experience with infectious diseases and the sources they drew on when simulating the specific pandemic scenario given in the study. Participants were more likely to draw on fictional sources when simulating an event with which they had little relevant direct experience. And the more relevant direct experience they had, the more likely they were to draw on those memories when simulating. These changes in source material were accompanied by changes in the content of the simulations. That is, simulations created after COVID-19 lockdown were rated as more vivid and emotional, and as eliciting a greater sense of reliving. One possible explanation comes from past findings that suggest that fictional memories are on average less vivid than personal memories (Yang et al., 2022); thus simulations prior to lockdown drew on a less vivid class on memories. However, memories of fiction/lived experience naturally co-varied with memory recency; by definition, memories of lived pandemic experiences were more recent and thus were expected to be more vivid on average. As described in the *Introduction*, prior work suggests that more vivid and recent memories tend to be more available, which in turn could increase their likelihood of affecting simulations (Bordala et al., 2025). Study 2 removed this confound by sampling a wider variety of scenarios, so that people were able to report memories of lived experience and fictional events from across their lives. Study 2 also removed the natural co-variance between individual experience and the experience of others, which likely inflated reliance on vicarious memories in Study 1 (as the college students' friends and

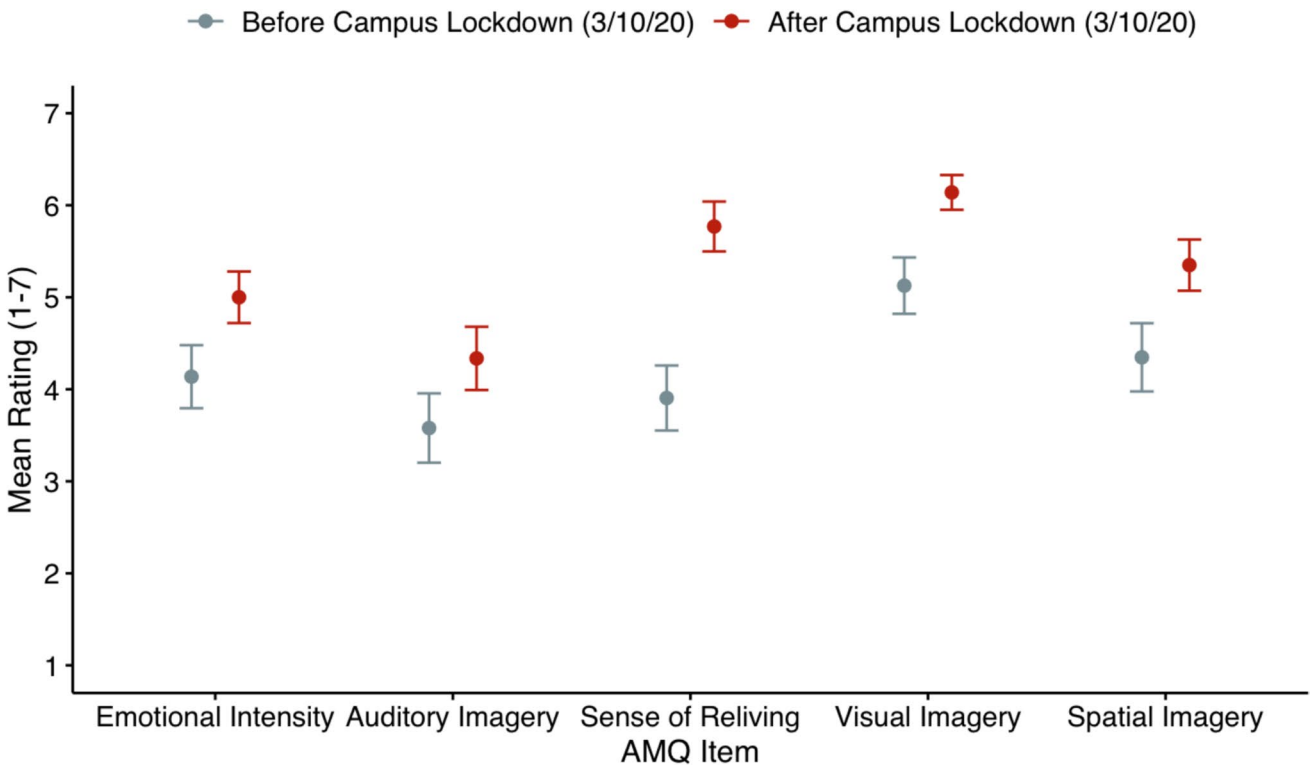


Fig. 2 Autobiographical Memory Questionnaire (AMQ) ratings of simulations by COVID-19 lockdown experience. Error bars represent 95% confidence intervals around the mean ratings

Table 2 **Proportion of** pandemic simulations drawing on each source type across levels of experience with infectious diseas**es**

Experience	Source type			
	Fiction	*Lived*	*Non-fiction*	*Vicarious*
1	90.32%	29.03%	51.61%	12.90%
2	95.00%	40.00%	55.00%	15.00%
3	77.78%	61.11%	50.00%	27.78%
4	68.00%	96.00%	48.00%	20.00%
5	44.19%	97.67%	51.16%	41.86%
6	32.39%	95.77%	43.66%	38.03%

Experience indicates the level of reported direct experience with the general event (i.e., infectious diseases). Proportions indicate the percentage of *simulations* that included each source. Given that participants could select multiple sources per simulation, these proportions are not mutually exclusive and thus do not sum to 1

Table 3 **Binomial** model using reported experience (1:6) to predict the probability that fiction is reported as source mater**ial (0,1) in Study 1**

Variables	Odds ratio [95% CI]	Coefficient (*SE*)	*p*-value
(Intercept)	29.33 [10.66, 95.71]	3.38 (0.56)	<.0001
Experience	0.50 [0.40, 0.62]	−0.69 (0.11)	<.0001

R^2 = 0.24 (Tjur),.19 (McFadden)..23 (Cox and Snell). Model $\chi 2(1)$ = 54.85, p <.0001. Proportion of correct predictions (PCP) =.72. AIC = 232.56, BIC = 239.24

Table 4 Binomial model using reported experience (1:6) to predict the probability that memories of lived experience are reported as source mat**erial (0, 1)**

Variables	Odds ratio (95% CI)	Coefficient (*SE*)	*p*-value
(Intercept)	0.13 (0.05, 0.30)	−2.05 (0.44)	<.0001
Experience	2.66 (2.07, 3.57)	0.98 (0.14)	<.0001

R^2 = 0.41 (Tjur),.38 (McFadden)..33 (Cox and Snell). Model $\chi 2(1)$ = 82.53, p <.0001. Proportion of correct predictions (PCP) =.86. AIC = 141.27, BIC = 147.95

family gained experience with lockdown at the same time as the students themselves).

More generally, Study 2 was a conceptual replication of Study 1. We generated scenarios that we predicted would elicit a range of prior experience and used pre-testing to select a subset for the study. Each participant in the main study rated their experience with the general class of event for each of the scenarios. These ratings were used to select four scenarios for simulation (two for which the individual had a lot of relevant experience and two for which the

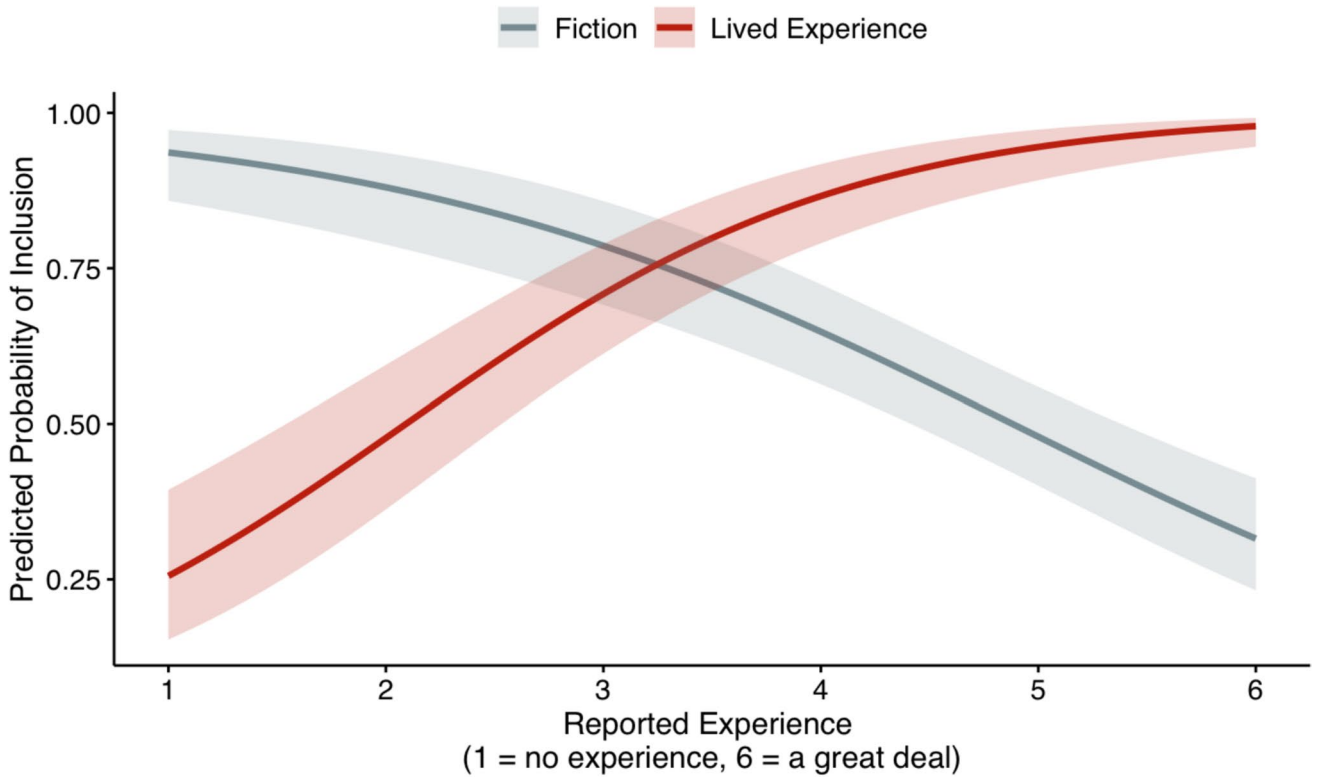


Fig. 3 Predicted probability of including fiction vs. lived experience as a function of reported experience in Study 1. Lines represent predicted probabilities from binomial logistic regression models estimating the likelihood of including fiction or lived experience based on reported experience with the general event (i.e., infectious diseases). Shaded bands indicate 95% confidence intervals

individual had less experience). Thus, we used a within-subjects design to explore the effects of lived experience on simulation content and source material, using a more diverse set of items. This study was pre-registered at osf.io/xrbyg.

Methods

Participants

A total of 248 participants (M_{age} = 33.78 [range 18–80], SD_{age} = 12.69 years; 135 women, 110 men, three gender unavailable) were recruited from Prolific in January 2021. We computed a power analysis for a logistic regression (not accounting for random effects) to determine sample size. To be conservative, we estimated an asymmetrical distribution of reporting fiction (.7,.3) and an odds ratio higher than in the previous study (OR =.75 here; OR =.55 in the previous study). To achieve.95 power at an alpha of.05, we estimated that we needed about 250 participants.

We pre-screened to select for workers who reported being fluent in English and currently resided in the UK, Canada, or the USA. Two additional participants were excluded from the analyses due to failing at least one attention check and providing open-ended responses that were off topic; exclusion decisions were made prior to data analysis. Participants were paid $3.50 USD to complete the task for an average hourly rate of $10.10 USD.

Materials

We generated an initial set of 28 scenarios that were expected to elicit varying levels of experience. Thirty-two Prolific participants (separate from the main study) used a 6-point scale to rate their lived experience with each of four to five randomly selected scenarios from this set (1 = no direct experience, 6 = a great deal of direct experience). We selected 16 items to use in the main study, choosing ones that elicited variation in experience across participants. We then created two versions of each item, one to elicit people's experience with the general class of events (e.g., "getting an MRI") and the other a specific scenario for simulation (e.g., "Imagine that you're at the hospital getting an MRI to help diagnose headaches"). The general prompts were used in the *selection* phase; our goal was for each person to simulate two events associated with greater lived experience and two events associated with less lived experience. In contrast,

the *simulation* prompts were specific; the specific scenarios were also used in the *experience assessment phase* at the end of the experiment.

The full set of 16 items (with both general and specific prompts) can be found in Appendix A.

Procedure

As in Study 1, each participant was asked to imagine and answer questions about four scenarios. Study 2 consisted of four phases: (1) *the selection phase*, (2) *the simulation phase*, (3) *the reporting phase*, and (4) *experience assessment phase*. The overall procedure of Study 2 was similar to Study 1, with the main difference being the addition of the selection phase.

In the *selection phase*, participants rated their personal experience (1 = no experience, 6 = a great deal of experience) with 16 general classes of events. For example, participants rated their experience with the general experience, "getting an MRI." Based on these initial ratings, we utilized internal logic within Qualtrics to assign four pseudo-randomly chosen scenarios to each participant for the subsequent simulation and reporting phases. Up to two of the scenarios were *high-experience items*, meaning the general events from these scenarios were rated a 4 or higher during the initial ratings. If participants rated more than two scenarios a 4 or higher, two were chosen randomly from the subset of high experience scenarios. Then, the *low-experience items* were randomly selected from the pool of scenarios that were rated 3 or lower during the initial ratings for a total of four items per participant. By using this pseudo-randomization method, most participants were assigned two scenarios they had relatively high experience with and two scenarios they had relatively low experience with.

In the *simulation phase*, participants were asked to imagine what they would do in the four specific scenarios they were assigned (two of which they reported higher levels of past experience, and two of which they reported lower levels of past experience). To help ensure that the online participants imagined the prompt, participants were also asked to describe what they imagined in at least two sentences.

In the *reporting phase*, participants answered questions about the source and phenomenology of each of their four simulations. This reporting phase was similar to that of Study 1: participants were reminded of what they wrote in the simulation phase for each scenario, reported on the sources of each simulation via open text response and a checklist, and rated the qualities of each simulation via AMQ questions relating to visual imagery, auditory imagery, reliving, emotional intensity, and spatial layout. Each of the reporting phase scenario blocks contained an attention check question. Questions for each scenario were presented in blocks, and each block was presented in a random order.

Finally, participants rated their *experience* with each of the four scenarios they had imagined; these ratings of experience were used as predictors in the analyses that follow. Participants answered a final attention check question, provided their age, and were offered the opportunity to provide comments or feedback. Other demographic information was downloaded from Prolific.

Statistical analyses

Analyses were completed in RStudio (2023) running R version 6.2.5. Generalized linear mixed-effects models were run using the "lme4" software package (Bates, Maechler, Bolker, & Walker, 2015). Significance for fixed effects was assessed using Satterthwaite approximations to degrees of freedom with the package "lmerTest" (Kuznetsova, Brockhoff, & Christensen, 2017).

Results

Selection of low and high experience events

People's responses in Phase 1 were used to select the events simulated in the experiment. Most participants (73.8%) were assigned two scenarios with which they had high general experience (rating ≥ 4) and two scenarios with relatively low experience (rating ≤ 3). The remaining participants were assigned one high-experience scenario and three low-experience scenarios (15.7%) or four low-experience scenarios (10.5%). Thus, our pseudo-randomization was fairly successful in assigning simulations that varied in how much past experience participants had to draw on. For the interested reader, the OSM shows the distribution of experience responses for each of the 16 items.

Sources of simulations

As in Study 1, participants could report multiple sources for each simulation: fiction, lived experience, non-fiction, and/or vicarious memories. These sources were not mutually exclusive, and proportions are summarized in Table 5. Our primary analyses investigated whether the use of fictional memories was more likely when participants reported having little relevant personal experience with the simulated scenario. The interested reader can find the analyses of vicarious and non-fictional memories in the OSM; to preview, personal experience with the specific scenarios was negatively related to reliance on non-fiction and unrelated to vicarious memories.

To replicate the findings from Study 1 that people draw on fictional sources when simulating in the absence of related lived experience, we ran two binomial logistic mixed-effects regressions predicting the use of fiction and lived experience

Table 5 Proportion of simulations including each source type across levels of reported experience with scenario

Experience (with scenario)	Source type			
	Fiction	*Lived*	*Non-fiction*	*Vicarious*
1	74.35	41.16%	24.35%	17.46%
2	70.87	70.08%	23.62%	27.56%
3	57.78	83.33%	22.22%	23.33%
4	44.55	83.64%	20.00%	20.91%
5	26.14	90.91%	12.50%	15.91%
6	33.96	94.34%	17.92%	13.21%

Experience indicates the level of reported first-hand experience with the specific scenario that was simulated. Percentages indicate the proportion of *simulations* that included each source. Given that participants could select multiple sources per simulation, these proportions are not mutually exclusive thus do not sum to 1

as source material during simulations, respectively. We first modeled whether participants reported using *memories of fiction* as source material for their simulations (1 = yes, 0 = no) with *reported experience* with the specific scenario from the imagination phase as the predictor and with *item* and *participant* as random effects. The mixed-effects model showed better fit than a fixed-effect model, as indicated by a lower AIC (1167.90 vs. 1204.10). AIC is a measure of model quality that balances goodness-of-fit with model complexity with lower values indicating a more parsimonious model. In this case, including random effects for "*participant*" and "*item*" improved the model's ability to account for variability in the data. Results showed that higher ratings for prior experience were significantly associated with a reduced likelihood of including fiction. Specifically, for each one-unit increase in reported experience, the odds of including fiction as a source decreased by 40% (OR = 0.60, 95% CI [0.54, 0.67], p <.0001, see Table 6).

Next, we analyzed whether participants reported using *lived experience* in their simulations (1 = yes, 0 = no) with reported experience with the scenario from the imagination phase as the predictor and item and participant as random effects. Reported experience was again a significant predictor, but in the opposite direction: for each one-unit increase in reported experience, the odds of including lived experience more than doubled (OR = 2.51, 95% CI [2.10, 3.00], p <.0001, see Table 7).

Taken together, these models replicate the results of Study 1 and are consistent with the hypothesis that people are more likely to recruit material from works of fiction into their simulations in the absence of relevant lived experience. As depicted in Fig. 4, the proportion of simulations incorporating memories of lived experience increased with reported experience (1–6), whereas the proportion of simulations including works of fiction declined.

Table 6 **L**ogistic mixed effects model regressing reported experience (1:6) on inclusion of fiction (0,**1) in Study 2**

	Odds ratio (95% CI)	Coefficient (*SE*)	*p*-value
(Intercept)	6.33 (4.14, 9.68)	1.84 (.22)	<.0001
Experience	0.60 (0.54, 0.67)	−0.42 (.06)	<.0001

ICC (adjusted) = 0.27, Conditional R^2 = 0.39, Marginal R^2 = 0.16. AIC = 1167.90, BIC = 1187.50. $N_{observations}$ = 985, N_{groups} = 247 (participant), 16 (item)

Phenomenology of simulations

The results of Study 1 suggested that episodic simulations become more vivid as experience with a given situation increases. We first sought to replicate this finding with the scenarios used in Study 2. To this end, we computed a multivariate linear regression predicting participants' ratings of AMQ items with reported experience (1–6) as the predictor and participant and scenario as random effects. This model predicted a 0.27 increase in phenomenological vivacity (1–7 scale) for each 1-unit increase in reported experience (1–6 scale; SE = 0.01, p <.0001; conditional R^2 = 0.51, marginal R^2 = 0.09, ICC = 0.46). Figure 5 provides visual confirmation that this positive relationship between reported experience and phenomenology ratings holds for individual AMQ items. This was further reinforced in follow-up models predicting individual AMQ items with reported experience (t-values > 3.72, p-values <.0001). These results suggest that the vivid simulations associated with lived experience are not attributable to the recency of lived experience (as was possible in Study 1).

Discussion

Across two studies, we examined the sources individuals drew on when simulating scenarios; critically, people drew on fictional events in the absence of lived experience. In Study 1, participants initially reported drawing on movies like Contagion when simulating a pandemic, however, after participants' campus closed due to the COVID-19 pandemic (i.e., lockdown), responses were drastically different with most responses drawing on lived experiences. In Study 2,

Table 7 Logistic mixed effects model regressing reported experience (1:6) on inclusion of lived experience (0,1) in Study 2

	Odds ratio (95% CI)	Coefficient (SE)	*p*-value
(Intercept)	0.28 (0.18, 0.44)	−1.27 (0.23)	<.0001
Experience	2.51 (2.10, 3.00)	0.92 (0.09)	<.0001

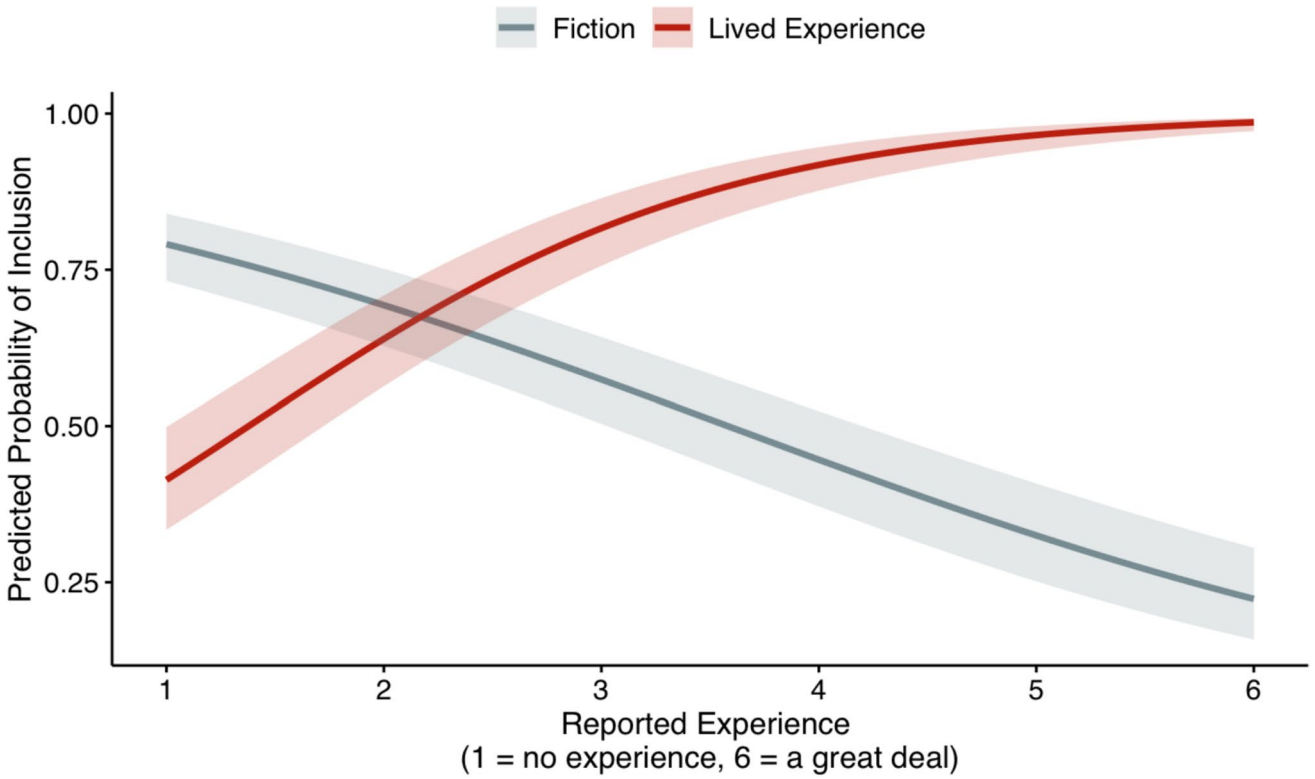
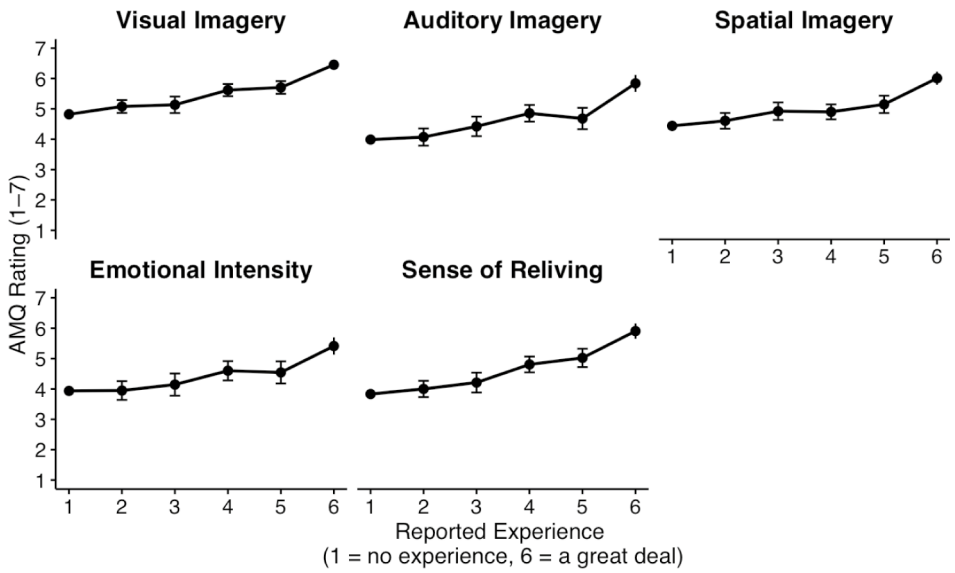


Fig. 4 Predicted probability of including fiction vs. lived experience as a function of reported experience in Study 2. Shaded areas indicate 95% confidence intervals. Visualized predictions are based on fixed effects only, averaged across participants and items



Note. Error bars represent 95% confidence intervals around the mean.

Fig. 5 Mean ratings of Autobiographical Memory Questionnaire (AMQ) items across levels of reported experience in Study 2. Error bars represent 95% confidence intervals around the mean

participants reported their personal experience with a series of general events and were subsequently presented with two scenarios they had some previous experience with and two prompts they had less experience with. Again, participants were more likely to draw on fictional sources when simulating scenarios they had less direct experience with. Overall, the results were consistent across the two studies despite the methodological differences; Study 1 held the scenario (i.e., a global pandemic) and possible fictional content constant but examined simulations as a function of people's experience, whereas Study 2 compared simulations for scenarios that each participant reported having more or less related experience stored in memory.

Across both studies, participants' simulations were rated as more vivid when lived experience was available as source material. In Study 1, this result was not surprising given that the participants retrieving lived experiences of the pandemic were drawing on very recently experienced events – or perhaps were even reflecting on their present experiences. While we did not directly measure the age of the source memories in Study 2, at a minimum we believe that there was likely less of a difference in the age of fictional and personal memories, as compared to Study 1. Regardless of the age of the memories, past work suggests that memories of fiction tend to be less vivid than memories of lived experience (Yang et al., 2022). The researchers note that this is not a hard and fast rule – fictional memories may be quite vivid if they are significant to the person, are frequently rehearsed, or are re-watched/re-read (see Silvosa et al., 2025) – but to the extent that vividness is an indicator of the strength or availability of memories, it suggests another possible reason why personal memories, when available, may be relied on over fictional memories.

We propose that the answer to *when* people draw on memories of fiction is largely explained by an individual's past personal experience relevant to the to-be-simulated event. In the absence of lived experience, individuals may rely on fiction. However, when both fictional and personal memories are present, it is likely advantageous to draw on what is known to be true (i.e., one's own experience) rather than trusting a fictional source that may be inconsistent with reality. Furthermore, memories of one's own experiences are more vivid in memory than fictional memories, likely making them more available and more useful as building blocks – consistent with recent modeling showing that priming affects which personal experiences are drawn upon when simulating the future (Bordalo et al., 2025). That is, increasing memory competition (by requiring retrieval of unrelated memories) makes people less likely to draw on their personal memories when estimating risk, highlighting how the (un) availability of memories influences whether they are used in simulations. One question for future research is whether the realism of a given piece of fiction influences the likelihood of events from that source being used to simulate the future; we hypothesize that an individual may be more likely to draw on stories set in familiar times and places than ones set in far-away worlds (in the same way that misinformation is more potent when embedded in stories set in more familiar contexts; see Rapp et al., 2014).

While we reported people's reliance on vicarious and non-fictional sources (see Tables 2 and 5), it should be noted that the present studies were not designed to capture reliance on these sources and we cannot draw strong conclusions. That is, while people reported using those sources, albeit at lower levels, it is not clear what the relationship is between one's personal experience and reliance on non-fictional and vicarious memories. There were two different patterns for vicarious memories across studies, likely because personal experience was directly correlated with other people's experiences in Study 1 but not in Study 2. Similarly, reliance on nonfiction sources was unrelated to personal experience in Study 1 but negatively related in Study 2; one possibility is that a more semantic simulation (e.g., a general prediction rather than imagining a specific event) would reveal a clearer pattern involving nonfiction sources. Future research will be needed to better understand when people rely on nonfictional and vicarious memories, perhaps using different scenarios or manipulating the availability of the target class of memories.

Our work depends on participants' ability to self-report the sources of their simulations, but we have no reason to believe that these reports were deceptive or systematically inaccurate. That is, while participants' sources may have been inferred rather than directly retrieved, there was no demand to report a particular source. Further, the switch in sources pre- versus mid-pandemic is consistent with actual events, suggesting that participants' source claims in Study 1 were likely accurate. Of course, people may have drawn on many different sources when simulating an event, whereas for simplicity we are independently considering the inclusion of personal versus fictional sources rather than considering the interaction between multiple sources when simulating an event. As noted earlier, many fictional sources incorporate elements of the real world – such as a movie scene set in Times Square, for example – and people may pull on multiple different sources when simulating a new event. Even the events in Study 2 that participants reported having experience with likely required drawing on more than one stored memory. That is, even if a participant had experiences with both MRIs and headaches, they may have had to combine multiple different personal memories to imagine the specific event described in the scenario. Future studies should further explore how different sources are pulled together when generating simulations.

Theoretically, our results are consistent with the claim that memory is an adaptive and constructive system (Schacter & Addis, 2007). That is, rather than

retrieving exact reproductions of the past, components of memory are used to construct and imagine possible futures (Suddendorf & Corballis, 2007) in order to navigate towards desired outcomes (Baumeister, Vohs, & Oettingen, 2016). The novel finding here is that memories of fiction are included in the building blocks that are used to reconstruct past memories and simulate future events. Our work supports the idea that, in addition to sharing some of the phenomenology of personal memories, memories of fiction also serve the same functions as memories of lived experience. Here we provide evidence that people could use fictional memories to simulate future action, with memories of fiction being used to generate simulations of specific scenarios. Much like how Green and Brock (2000) propose that transportation, the capacity to mentally transport into narrative worlds, can affect our beliefs and evaluations regardless of whether a story is labeled as fact or fiction, we posit that people can seamlessly go between drawing from memories of lived experience and fiction. This supports the idea that the line between fact and fiction can be blurred, especially when memories of fiction and lived experience share similar phenomenological qualities and serve the same functions.

Overall, extending what we can imagine beyond the narrow band of our own lived experiences is useful. This claim aligns nicely with the goals of many children's storybooks that share tales highlighting desired societal behaviors and punishing transgressions. This ability to encode, store, retrieve, and recombine elements of second-hand episodic memories to generate new solutions and simulate new possibilities is likely a unique property of human cognitive systems. This computationally costly ability holds great adaptive value in terms of population-level survival and reproduction (Suddendorf & Corballis, 2007). However, our willingness to draw on event memories in constructing simulations, collapsing over truth status, holds value and significance in our modern world too.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.3758/s13421-025-01801-1.

Authors' contributions B.W.Y. conceived the original idea and carried out the experiments. J.S.P. and E.J.M. ran the analyses and wrote the manuscript. F.D. and E.J.M helped supervise the project.

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Data availability The materials and data for Studies 1 and 2 are openly available at the project's Open Science Framework (OSF) page (osf.io/hxqck). The pre-registration for Study 2 is also available on the OSF (https://osf.io/xrbyg).

Code availability The code for analyses are available on the OSF (osf.io/hxqck).

Declarations

Ethics approval This study was approved by the Duke University Campus Institutional Review Board (IRB #2017-0147).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication The authors affirm that human research participants provided informed consent for publication of their de-identified data.

Conflicts of interest The authors have no conflicts of interests to disclose.

Open Practices Statement The materials, data, and code for Studies 1 and 2 are openly available at the project's Open Science Framework (OSF) page (osf.io/hxqck). The pre-registration for Study 2 is also available on the OSF (https://osf.io/xrbyg).

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